

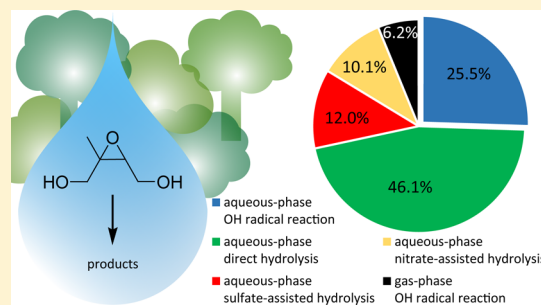
Aqueous-Phase Oxidation of *cis*- β -Isoprene Epoxydiol by Hydroxyl Radicals and Its Impact on Atmospheric Isoprene Processing

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Supporting Information

ABSTRACT: Isoprene epoxydiols are well-known intermediates of the isoprene oxidation in the gas phase under pristine conditions. According to their Henry's Law constants, they are well water-soluble and will undergo a phase transfer to the tropospheric aqueous phase to be further processed there. In the present study, the aqueous-phase oxidation by hydroxyl radicals (OH) of the isomer *cis*- β -isoprene epoxydiol (*cis*- β -IEPOX) was investigated with regards to its kinetics and reaction product distribution. For the reaction of *cis*- β -IEPOX with hydroxyl radicals, a second-order rate constant of $(1.4 \pm 0.1) \times 10^9 \text{ L mol}^{-1} \text{ s}^{-1}$ was determined at 298 K. Short chain carbonyls, such as glycolaldehyde with a maximum molar fraction of 6.9%, hydroxyacetone (7.4%), methylglyoxal (10.3%), and glyoxal (6.6%), as well as acetic acid (13.9%), were observed as major oxidation products. A total molar recovery of 75% was obtained in the oxidation experiments. The overall contribution of the aqueous-phase oxidation of *cis*- β -IEPOX was estimated to range from 29.7 to 21.3% to the atmospheric processing depending on the scenario.

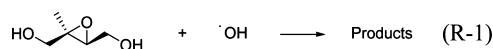


INTRODUCTION

Isoprene is the most abundant nonmethane biogenic volatile organic compound (VOC) with a global source strength of $594 \pm 34 \text{ Tg a}^{-1}$ accounting for more than 65% of the total carbon mass of biogenic emitted compounds.¹ Once released into the atmosphere by a variety of plants,² isoprene is rapidly oxidized by the abundant gas-phase oxidants, hydroxyl radicals ($k_{\text{OH}+\text{isoprene}} = 9.99 \times 10^{-11} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$), ozone ($k_{\text{O}_3+\text{isoprene}} = 1.27 \times 10^{-17} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$), and nitrate radicals ($k_{\text{NO}_3+\text{isoprene}} = 6.52 \times 10^{-13} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$).³ This oxidative gas-phase chemistry leads to a vast number of oxygenated VOCs,^{4–22} which has recently been reviewed by Wennberg et al. comprehensively,²³ focusing on pristine conditions where the chemistry is HO_x -dominated isoprene hydroxyhydroperoxides (ISOPOOH) are formed.¹⁹ During the ongoing OH oxidation, the main product from ISOPOOH is IEPOX with a yield of 71–79% depending on the ISOPOOH isomer.^{19,22} Further oxidation of IEPOX is poorly described in the literature, probably because atmospheric IEPOX processing is mainly believed to occur through particle chemistry.²³ This assumption is based on the good water-solubility of IEPOX and its ability to undergo reactive uptake on acidic sulfate particles. The Henry's law coefficients given by the literature and estimation methods are in the range of $10^8 \text{ mol L}^{-1} \text{ atm}^{-1}$. In detail, the estimation of the Henry's law constant of IEPOX resulted in a value of $2.7 \times 10^6 \text{ mol L}^{-1} \text{ atm}^{-1}$ using the bond estimation method and $2.9 \times 10^{10} \text{ mol L}^{-1} \text{ atm}^{-1}$ from the group estimation method.²⁴ Eddingsaas et al.²⁵ published an estimated Henry's law coefficient of $1.3 \times 10^8 \text{ mol L}^{-1} \text{ atm}^{-1}$ using a calibrated HENRYWIN estimation.²⁴

Supporting both these estimation methods, Gaston et al.²⁶ derived an average Henry's law constant of $1.7 \times 10^8 \text{ mol L}^{-1} \text{ atm}^{-1}$ for sulfate-containing aerosols using a resistor model to model their experimental data.²⁶ Based on the estimated values, the IEPOX partitions with an aqueous-phase fraction up to 99% (cf. Figure S1).

The initial aqueous-phase chemistry of IEPOX is thought to be dominated by the acid-catalyzed ring-opening of the epoxide ring and subsequent reactions.^{25–30} Moreover, the direct aqueous-phase radical chemistry of IEPOX is competing to the hydrolysis chemistry, especially in diluted systems such as cloud droplets where the pH is more neutral ($\text{pH} = 4\text{--}6$) and the nucleophile concentration is low ($c \approx 10^{-5} \text{ mol L}^{-1}$) compared to aerosols. In the present study, the aqueous-phase oxidation of *cis*- β -isoprene epoxydiol (*cis*- β -IEPOX) by hydroxyl radicals (cf. R-1) will be investigated with regards



to its kinetics by using laser flash photolysis. Mechanistic investigations concerning the formed products will be performed by applying different mass spectrometric techniques as well as capillary electrophoresis. These experimental findings will be supported by ab initio calculations, and the overall impact on the branching of the atmospheric *cis*- β -IEPOX reactions will be assessed by flux analysis.

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METHODS

Materials and Chemicals. The following chemicals were used as purchased without further purification: hydrogen peroxide (Chemsolute, 30 wt % in H₂O), KSCN (Chemsolute, >99%), O-(2,3,4,5,6-pentafluorobenzyl) hydroxylamine hydrochloride (Sigma-Aldrich, 99.0%), 4-(trifluoromethyl)-benzaldehyde (Sigma-Aldrich, 98.0%) hydrochloric acid (Merck, 37 wt %), dichloromethane (Sigma-Aldrich, 99.9%), hydroxyacetone (Alfa Aesar, 95%), glycolaldehyde dimer (Sigma-Aldrich, melts between 80 and 90 °C), glyoxal (Sigma-Aldrich, 40 wt % in water), methylglyoxal (Sigma-Aldrich, 40 wt % in water), sodium acetate (Sigma-Aldrich, puriss. p.a.), tris(hydroxymethyl)aminomethane (Merck, ≥99.8%), 5-sulfosalicylic acid (Merck, ≥99%), acetonitrile (Fisher, Optima LC/MS grade), and formic acid (Merck, LiChropur, 98–100%).

Cis-β-IEPOX was synthesized following the procedure originally published by Zhang et al.³¹ and modified by Bates et al.³² A ¹H NMR spectrum of the synthesized compound is shown in the [Supporting Information](#) (cf. Figure S18).

Aqueous solutions were freshly prepared with ultrapure water (18.2 MΩ cm, TOC < 10 ppb) prior to use.

Kinetics. The OH radical kinetics were measured using a laser flash photolysis—laser long path absorption setup described in detail previously.^{33–36} The second-order rate constants were measured using a relative competition kinetics technique, firstly described by Behar et al.,³⁷ with potassium thiocyanate as a competitor to avoid the influence of possibly co-absorbing species, such as peroxy radicals. Formed hydroxyl radicals ($c_{\text{max}} = 3.3 \times 10^{-7}$ mol L⁻¹) react with dissolved thiocyanate under the formation of dithiocyanate radical anions ((SCN)₂•⁻) which strongly absorb in the blue visible light region ($\epsilon_{\text{max}}(478 \text{ nm}) = 7539 \text{ L mol}^{-1} \text{ cm}^{-1}$).³⁸ Upon addition of *cis*-β-IEPOX, the initial concentration of (SCN)₂•⁻ decreases and subsequently the second-order rate constant of the reaction of *cis*-β-IEPOX with hydroxyl radicals is calculated according a literature method³⁹ (I)

$$\frac{A_{((\text{SCN})_2\bullet^-)_0}}{A_{((\text{SCN})_2\bullet^-)_x}} = \frac{k_{2\text{nd}}[\text{IEPOX}]}{k_{\text{REF}}[\text{SCN}^-]} + 1 \quad (\text{I})$$

$A_{((\text{SCN})_2\bullet^-)_0}$ is the absorbance of the dithiocyanate radical anion without the addition of *cis*-β-IEPOX, $A_{((\text{SCN})_2\bullet^-)_x}$ is the absorbance of the dithiocyanate radical anion with the addition of *cis*-β-IEPOX, [SCN⁻] is the concentration of dissolved thiocyanate, [*cis*-β-IEPOX] is the concentration of dissolved *cis*-β-IEPOX, $k_{2\text{nd}}$ is the desired second-order rate constant of the reaction of *cis*-β-IEPOX with hydroxyl radicals, and k_{REF} is the second-order rate constant of the reference system. For k_{REF} , the temperature-dependent value given by Zhu et al.⁴⁰ was used.

Mechanistic Investigation. Mechanistic investigations were performed in an aqueous-phase photoreactor which consists of a double-walled glass vessel ($V = 300 \text{ mL}$) tightly closed with a quartz glass window. As a light source, a xenon short arc lamp (450 W, LSB541, LOT Quantum Design) was used. A set of optical filters (water, 5 mm WG295 and 5 mm WG305) was applied to avoid IR irradiation and UV below 290 nm to mimic the actinic spectrum (cf. [Figure S2](#)). The temperature was kept at 298 K over the experiment time of 240 min. During the experiments, the following initial conditions were applied: $[\text{H}_2\text{O}_2] = 3 \times 10^{-3} \text{ mol L}^{-1}$, [*cis*-β-

IEPOX] = $1 \times 10^{-4} \text{ mol L}^{-1}$, and pH = 6.0 ± 0.1 . These conditions led to a steady-state concentration of hydroxyl radicals of $6 \times 10^{-14} \text{ mol L}^{-1}$. Samples of the reactor solution for each different analysis procedure described in the following were taken at intervals of 15 min for 3 h and afterward at 30 min over the remaining experiment time. Each experiment was performed twice.

UPLC Analysis. The analysis of *cis*-β-IEPOX and possibly formed oxidation, functionalization, and accretion products was performed on an ultra-high performance liquid chromatography system (UPLC) coupled to an Q-Exactive Hybrid Quadrupole-Orbitrap mass spectrometer. For this analysis, 200 μL of the reactor solution was sampled and 20 μL of catalase solution ($\gamma = 4 \text{ mg mL}^{-1}$) was added to avoid possibly further dark oxidation by hydrogen peroxide. Then, 200 μL of acetonitrile was added to the sample. For the quantification of *cis*-β-IEPOX, calibration standards in the range of 1×10^{-6} to $1 \times 10^{-4} \text{ mol L}^{-1}$ were used. Samples prepared following the described protocol were subsequently analyzed applying the following instrumental settings. The UPLC instrument was equipped with a Waters Acquity HSS T3 column (1.8 μm, 100 × 2.1 mm) operated with (solvent A) 0.1% formic acid in water and (solvent B) 0.1% formic acid in acetonitrile as eluents with a total flow of $300 \mu\text{L min}^{-1}$ at 40 °C. As a gradient system, the following elution program was used: 1 min 95% A/5% B followed by a linear gradient down to 2% A/98% B within 15 min followed by flushing to the starting conditions within 5 min. The mass spectrometer was equipped with a heated electrospray ionization source operating with the following parameters: 3.0 kV spray voltage, sheath gas flow of 50, auxiliary gas flow of 10, 280 °C capillary temperature, and 250 °C probe heater temperature. In a range of 50–750 m/z , masses were recorded with a resolving power of 70 000 at m/z 200. The mass axis calibration was performed using a $2 \times 10^{-3} \text{ mol L}^{-1}$ solution of sodium acetate.

GC–MS Analysis. Samples of 2 mL were taken at the given intervals to monitor the production of carbonyl group-bearing compounds. A solution of catalase (100 μL, $\gamma = 4 \text{ mg mL}^{-1}$) was added to the sample to avoid ongoing oxidation reactions due to the decomposition of excess hydrogen peroxide by the catalase. An internal standard of 100 μL 4-(trifluoromethyl)-benzaldehyde ($c = 6.65 \times 10^{-5} \text{ mol L}^{-1}$) was added to each sample. Afterward, an aqueous solution of O-(2,3,4,5,6-pentafluorobenzyl)hydroxylamine hydrochloride ($V = 200 \mu\text{L}$, $\gamma = 5 \text{ mg mL}^{-1}$) was added to each sample, and the samples were derivatized over 18 h at 20 °C. Calibration standards were prepared with the same protocol to quantify the formed products. Afterward, the samples were adjusted to pH = 1.0 ± 0.1 with hydrochloric acid (37 wt %), and the oxime derivatives were extracted by adding 250 μL of 2,2,4-trimethylpentane and shaking with an orbital shaker for 30 min. The organic layer was separated for the analysis of the derivatized carbonyl compounds on a gas chromatography system (Agilent 7890B) with a multimode inlet coupled to an electron impact mass spectrometer (Agilent 5977B) without further drying. The organic extract (10 μL) was injected into the gas chromatography–mass spectrometry (GC–MS) system. The separation was performed on an Agilent HP-5MS UI (30 m × 0.25 mm, 0.25 μm) under the following conditions: flow 1.2 mL min^{-1} , inlet temperature program: 70 °C isothermal for 0.41 min and increased to 325 °C with 600 K min⁻¹, temperature program: 50 °C for 2 min, increased to

210 °C with 10 K min⁻¹, and held for 7 min followed by a 10 min post run at 320 °C.

CE-UV Analysis. For capillary electrophoresis with ultraviolet light detector (CE-UV) analysis, 100 μL of the reactor solution was taken and 10 μL of the catalase solution was added to stop further oxidation by hydrogen peroxide. The analysis of small acids was performed on an Agilent 7100 CE-UV instrument equipped with a HPCE standard capillary (G1600-62311, 75 μm inner diameter, 72 cm length, Agilent Technologies). The operating high voltage was kept at 30 kV with a current of 50 μA. As a background analyte, a solution of 5-sulfosalicylic acid (2×10^{-3} mol L⁻¹) and tris-(hydroxymethyl)aminomethane (8×10^{-3} mol L⁻¹) was used. A hydrodynamic injection mode (35 mbar for 10 s) was used to introduce the taken sample. The quantification of the analyte was done by measuring the calibration standards of the compound with the same instrumental setting.

Bond Dissociation Energy Calculations. The bond dissociation energy (BDE) calculations were performed using GAMESS.^{41,42} Except for this, the calculations were performed as previously described by our group.⁴³ In brief, the geometry optimization of the molecular structure and the possibly formed radical structures were analyzed at the M06-2X level with the 6-311+G(d,p) basis set, and for the calculation of the zero-point energies and electronic energies, the basis set was changed to 6-311++G(3df,2p). Subsequently, the BDEs were calculated according to He et al.⁴³

RESULTS AND DISCUSSION

Kinetics and Activation Parameters. The second-order rate constants of the reaction of *cis*-β-IEPOX with hydroxyl radicals were determined at pH = 6.0 ± 0.1 in a temperature range from 278 to 318 K. The obtained temperature-dependent second-order rate constants are shown in Figure 1.

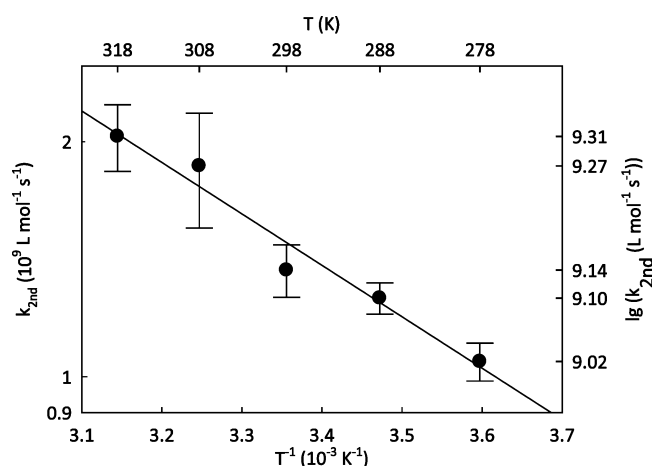


Figure 1. Arrhenius plot for the reaction of *cis*-β-IEPOX with hydroxyl radicals.

Experimental second-order rate constants at the different temperatures are k_{2nd} (278 K) = $(1.1 \pm 0.1) \times 10^9$ L mol⁻¹ s⁻¹, k_{2nd} (288 K) = $(1.3 \pm 0.1) \times 10^9$ L mol⁻¹ s⁻¹, k_{2nd} (298 K) = $(1.4 \pm 0.1) \times 10^9$ L mol⁻¹ s⁻¹, k_{2nd} (308 K) = $(1.9 \pm 0.3) \times 10^9$ L mol⁻¹ s⁻¹, and k_{2nd} (318 K) = $(2.0 \pm 0.2) \times 10^9$ L mol⁻¹ s⁻¹. The errors are statistical errors of the linear regression (least square method) of $\frac{A_{((SCN)_2^{\bullet-})_0}}{A_{((SCN)_2^{\bullet-})_X}}$ as a function of $[cis\text{-}\beta\text{-IEPOX}]/[SCN^{\bullet-}]$ as needed for the calculations according to (I) multiplied by the appropriate quantile of the Student's *t*-distribution for 95% confidence interval. To the author's knowledge, there is no comparable aqueous-phase kinetic data in the literature available, and even the data on structural similar compounds is sparse. There is one publication by Anbar et al.⁴⁴ on the reaction kinetics of epoxides with hydroxyl radicals. These authors reported reaction rate constants of 3.9×10^7 L mol⁻¹ s⁻¹ for 1,2-epoxyethane as the basis molecule, 4.5×10^8 L mol⁻¹ s⁻¹ for its substituted derivative 1,2-epoxybutane, and finally, 2.7×10^8 L mol⁻¹ s⁻¹ for 2,3-epoxypropanol. In the case of 1,2-epoxybutane, the CH₂ and CH₃ groups are the additional reactive sides to the remaining ring hydrogen. In case of 2,3-epoxypropanol, there is only the CH₂ group in the α-position to the ring and the OH group causing an increase of 2.3×10^8 L mol⁻¹ s⁻¹. In comparison, in *cis*-β-IEPOX, there are four major contributing pathways, the H abstraction at the CH₂ groups in the α-position to the ring, the methyl group attached to the ring carbon, and a tertiary hydrogen attached to the second ring carbon. Therefore, the higher reaction rate constant of *cis*-β-IEPOX reported in this study is caused by the higher number of reactive sites but still comparable to the less substituted epoxides investigated by Anbar et al.⁴⁴ In contrast to this conclusion, calculations using the structure activity relationship method after Monod and Doussin^{45,46} resulted in an overall second order rate constant of $k_{overall} = 2.4 \times 10^9$ L mol⁻¹ s⁻¹ (see the Supporting Information for the calculation) overestimating our experimental value for 298 K by a factor of 1.7. The overestimation may be caused by missing factors for the ring structure. Whenever a factor for OR was required, it was substituted by OH. Nevertheless, the estimation is still in a range of factor 2 which tends to be sufficient for such estimation methods.

The linear regression of the experimental temperature-dependent data of the Arrhenius plot (Figure 1) leads to the following expression

$$k(T) = (2.4 \pm 0.1) \times 10^{11} \times \exp\left(\frac{-1520 \pm 170 \text{ K}}{T}\right) \text{ L mol}^{-1} \text{ s}^{-1} \quad (\text{II})$$

From this equation, the following thermochemical properties (Table 1) are derived according to the calculations previously described^{47,48}

Table 1. Obtained Arrhenius Parameters and Derived Thermochemical Properties for the Reaction of *cis*-β-IEPOX with Hydroxyl Radicals

activation parameter	experimental value
activation energy	12.6 ± 1.4 kJ mol ⁻¹
A factor	$(2.4 \pm 0.1) \times 10^{11}$ L mol ⁻¹ s ⁻¹
activation enthalpy	10.1 ± 1.4 kJ mol ⁻¹
activation entropy	$-(35.3 \pm 0.9)$ J mol ⁻¹ K ⁻¹
Gibbs activation energy	20.7 ± 3.4 kJ mol ⁻¹

The low activation energy leads to the assumption that the reaction mechanism of the reaction of *cis*-β-IEPOX with hydroxyl radicals has no significant chemical reaction barrier in line with the fact that this activation energy can already be caused by only the temperature-dependent change of the solvent viscosity alone which would correspond to an activation energy of around 15 kJ mol⁻¹.⁴⁹ The negative

activation entropy indicates a transition state with a higher degree of order compared to the educts. Besides the kinetics, a major task is always the elucidation of the formed products.

Product Studies and Mechanism. Product studies of the reaction of *cis*- β -IEPOX with hydroxyl radicals were performed to elucidate its product distribution, its impact on the oxidation budget, and its contribution to the formation of SOA precursors. The reactions were carried out in a pure aqueous system without any addition of influencing organics or inorganics, such as sulfuric acid, to ensure that the isolated hydroxyl radical-driven oxidation yields the detected products. Over the experiment time of 4 h, about 56% of the initial *cis*- β -IEPOX concentration is processed to its oxidation products (cf. Figures 2 and S3) by OH with a steady-state concentration of $6 \times 10^{-14} \text{ L mol}^{-1} \text{ s}^{-1}$ similar to the OH concentrations applied by Otto et al.³⁴

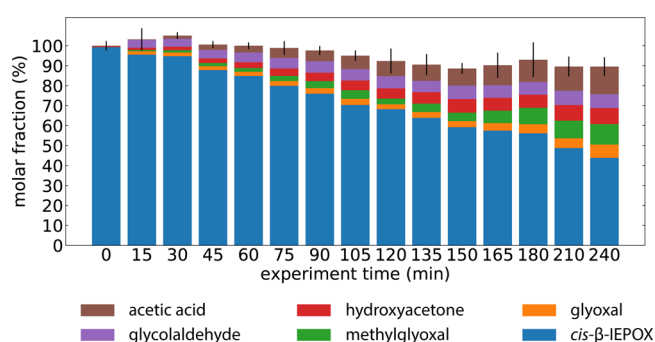


Figure 2. Time-dependent molar fractions of observed oxidation products for the OH-driven oxidation of *cis*- β -IEPOX.

The experimental findings are supported by density functional theory calculations on the bond dissociation energies. The favored positions for a H abstraction in the molecule are the hydrogens at C4 and C5 positions (Figure 3) with bond dissociation energies of 367 and 388 kJ mol^{-1} , respectively, also indicated by the SAR calculations after Monod and Doussin^{45,46} (Figure 3, cf. Supporting Information).

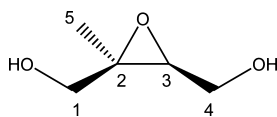


Figure 3. Assigned structure of *cis*- β -IEPOX.

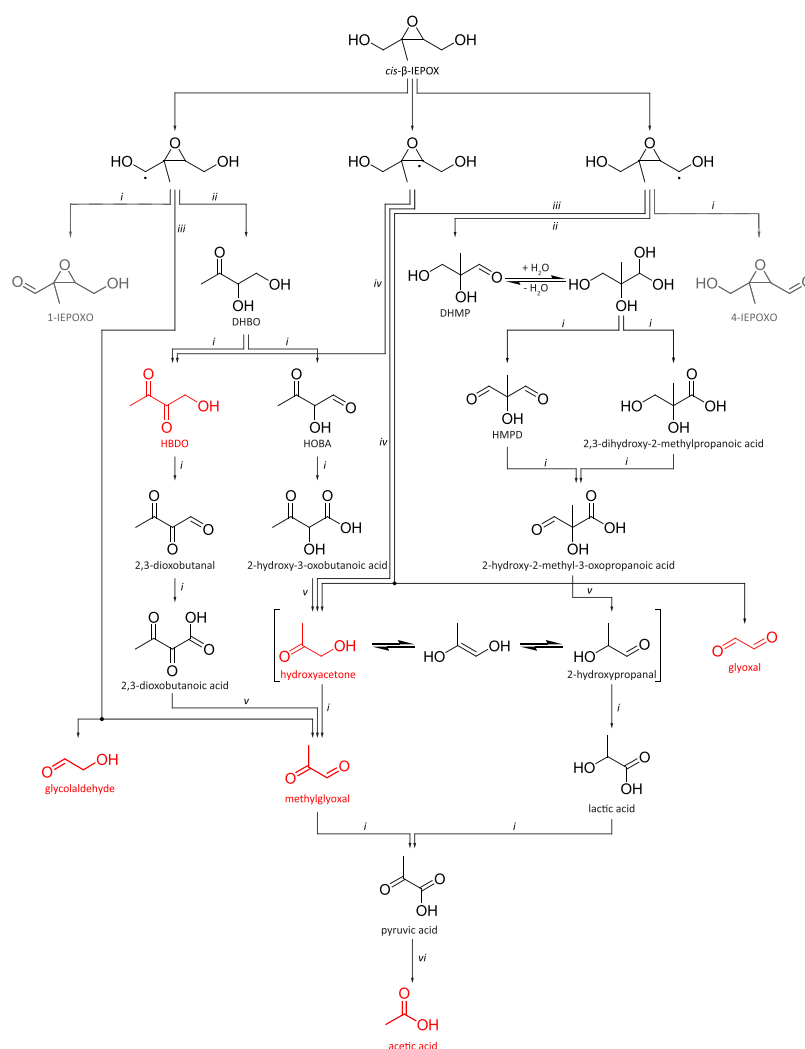
Following the suggested gas-phase mechanism,²³ based on Jacobs et al.⁵⁰ and Bates et al.,^{32,51} the H abstraction should occur at the C4 position resulting in the formation of glyoxal and hydroxyacetone, as reported by both groups mentioned above, as well as 2,3-dihydroxy-2-methylpropanal (DHMP) with all three species as stable “closed-shell” products as suggested by Bates et al.⁵¹ In contrast to the previous study, Jacobs et al.⁵⁰ have not detected products with the molar mass of DHMP but suggest the occurrence of a C₅ dihydroxydicarbonyl species. In the present aqueous-phase study, we observed glyoxal with a maximum molar fraction of 6.6% (cf. Figures 2 and S4) and hydroxyacetone with a maximum molar fraction of 7.4% (cf. Figures 2 and S5), respectively. Surprisingly, DHMP is not detected in the experiments, but it could be a transient intermediate, with a concentration not

exceeding our detection limit corresponding to an upper limit of concentration of $[\text{DHMP}]_{\text{max}} = 1 \times 10^{-5} \text{ mol L}^{-1}$. The suggested C₅ dihydroxydicarbonyl species was also not observed by our aqueous-phase study.

The further oxidation of DHMP in the diluted aqueous phase could result in the detected species as reported by Otto et al.³⁴ and also is depicted in Scheme 1. The H abstraction at the C5 position potentially yields the stable products methylglyoxal, glycolaldehyde, and 3,4-dihydroxybutanone (DHBO) in the gas phase. For the aqueous-phase oxidation, maximum molar fractions of 10.3 and 6.9% were detected for methylglyoxal (cf. Figures 2 and S6) and glycolaldehyde (cf. Figures 2 and S7). Potentially formed DHBO was not detected as already explained for DHMP above, so that the peak DHBO concentration may be below $[\text{DHBO}]_{\text{max}} = 1 \times 10^{-5} \text{ mol L}^{-1}$ if it is formed. The fact that DHBO and DHMP are not detected by this study could suggest that these compounds might be formed in the gas phase only. However, Jacobs et al.⁵⁰ have not detected a product with this mass. Both H abstraction pathways can potentially lead to *cis*- β -IEPOX derivatives (1-IEPOXO and 4-IEPOXO), as suggested for the gas-phase,^{23,32,50,51} where one hydroxy group is oxidized to an aldehyde group. Hence, there is no indication of their formation in the aqueous phase although it is expected. Therefore, it is shown in grey and with dashed reaction arrows in the following Scheme 1 to keep it complete. The H abstraction at the C₃ and C1 position is hardly probable because of their high BDEs of 413 and 404 kJ mol^{-1} . The high molar fraction of acetic acid of 13.9% is potentially formed via the intermediates hydroxyacetone and methylglyoxal as well as via the DHMP and DHBO pathways.³⁴ In all these pathways, hydroxyacetone is oxidized to methylglyoxal and then pyruvic acid which then is expected to break down to acetic acid (cf. Figures 2 and S8). Nevertheless, the increase of the acetic acid concentration even at smaller experiment times leads to the assumption of the occurrence of a direct formation pathway. Based on these experimental findings and our previous publication,³⁴ as well as the published gas-phase oxidation mechanisms,^{23,50,51} we postulate the following comprehensive aqueous-phase oxidation mechanism (cf. Scheme 1).

The three potential initial reactions shown in Scheme 1 are H-abstraction reactions at the most probable positions. Followed by four different reactions: (i) initial H abstraction followed by an HO₂ elimination, (ii) formation of the peroxy radical and a tentative H-shift to release CO and a hydroxyl radical forming DHBO and DHMP, (iii) intermediate formation of peroxy radicals followed by the postulated recombination pathways⁵² and the fragmentation of the molecule because of its strong functionalization and missing reactions sites, and (iv) radical fragmentation of the subsequent formed alkoxy radical [as for pathway (iii)] to a “closed-shell” product and a radical fragment. It should be noted that the detailed mechanism of the pathway (iii) is highly unclear although they are evidenced by the abundance of the formed products. Further ongoing chemistry of DHBO and DHMP is described in detail by our previous publication³⁴ where the pathway (v) is a decarboxylation and (vi) the oxidation by hydrogen peroxide. In general, this is the first attempt to propose an aqueous-phase oxidation mechanism of *cis*- β -IEPOX in line with the experimental findings of the present study and the former results from the literature on the respective gas-phase processes.

Scheme 1. Schematic of the Postulated Aqueous-Phase Oxidation Mechanism of *cis*- β -IEPOX Based on Experimental Findings and Our Previous Publication by Otto et al.³⁴ As Well As the Reviewed Gas-Phase Oxidation Mechanisms^{23,50,51}



Compounds which lead to mass to charge ratios not directly assignable are detected in the GC–MS analysis such as signals at a retention time of 10.0 and 10.8 min with a mass to charge ratio (m/z) of 281 or at 10.9 and 11.0 min with a mass to charge ratio of 267 of the biggest abundant fragment (cf. Figure S9). Furthermore, compounds with m/z 492, 490, 478, and 460 are observed (cf. Figures S10–S13) and is thought to be C_4 dicarbonyl species indicated by their even m/z and their retention time. That is, in the range of 21 to 23 min which usually is the retention time range of dicarbonyls in our GC–MS method. The $m/z = 492$ could indicate the occurrence of either 1-hydroxy-2,3-butadione (HBDO), 2-hydroxy-3-oxobutanal, or 2-hydroxy-2-methylpropanedial, as well as a mixture of these compounds all having the same molar mass and even the same m/z which are described as possibly formed oxidation products of DHBO and DHMP.³⁴ For the three other mass-to-charge ratios, a compound suggestion cannot easily be given. Under the assumptions described above and if these m/z ratios represent the molecular ion peaks, the parent compounds have molar masses of 100, 88, and 70 g mol^{-1} . For the compound having a molar mass of 100 g mol^{-1} , the most probable sum formulas are $C_5H_8O_2$ and $C_4H_4O_3$ both bearing at least two carbonyl groups. The best guess for the compound ($M = 88 \text{ g mol}^{-1}$) may be a C_3 dicarbonyl ($C_3H_4O_3$) as a product with

this molar mass is also suggested for the gas phase.⁵⁰ For the possible $M = 70 \text{ g mol}^{-1}$ compound, no reasonable guess can be given.

Despite these unassigned m/z , the determined concentrations of the performed experiments address at least 75% of the theoretical total molar fraction. The co-production of methylglyoxal and glycolaldehyde as well as the co-production of glyoxal and hydroxyacetone is considered. In the following, the experimental data is fitted with a chemical model to comprehend the branching of the potential pathways.

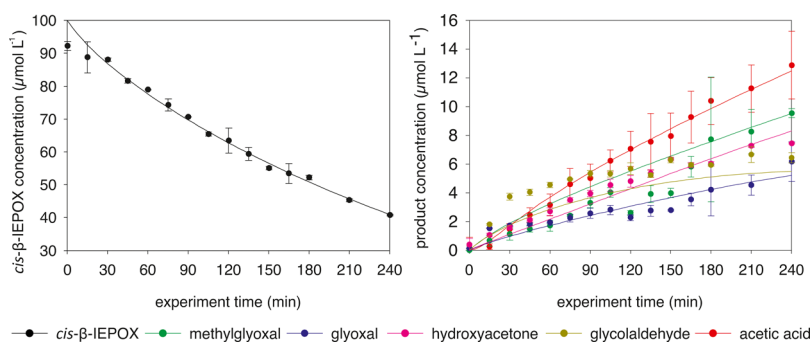
Experiment Modeling. Based on the postulated mechanism in Scheme 1 of the possibly occurring chemistry described in the previous section, a model mechanism is set up to reproduce the concentration–time profiles. In the model, the three initial OH radical attacks and the reaction to the first generation of stable reaction products are described by seven reactions (Table 2, entries 8 to 14) to describe the six solid pathways and the one summarizing both the pathways to the IEPOXO species in Scheme 1. The mechanism of the subsequent OH oxidation reactions (cf. Table 2, entries 15 to 21) are described in detail by our previous publication.³⁴ The formation of acetic acid is thought to be mainly caused by the oxidation reactions of α -hydroxy- and α -oxo-carbonyls with

Table 2. Reactions and Known Rate Constants (Standard Case) Considered in the COPASI Model and Modeled Reactions with Their Respective Branching Ratios (Bold Case)

entry	reaction	rate constants and estimated pathway contributions	refs
1	$\text{H}_2\text{O}_2 \rightarrow \text{OH} + \text{OH}$	$2.8 \times 10^{-6} \text{ s}^{-1}$	34
2	$\text{OH} + \text{OH} \rightarrow \text{H}_2\text{O}_2$	$3.6 \times 10^9 \text{ L mol}^{-1} \text{ s}^{-1}$	55
3	$\text{OH} + \text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{HO}_2$	$2.7 \times 10^7 \text{ L mol}^{-1} \text{ s}^{-1}$	56
4	$\text{OH} + \text{HO}_2 \rightarrow \text{H}_2\text{O} + \text{O}_2$	$6.0 \times 10^9 \text{ L mol}^{-1} \text{ s}^{-1}$	55
5	$\text{OH} + \text{O}_2^- \rightarrow \text{OH}^- + \text{O}_2$	$1.1 \times 10^{10} \text{ L mol}^{-1} \text{ s}^{-1}$	57
6	$\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$	$k_{\text{forward}} = 0.0014 \text{ s}^{-1}$, $k_{\text{backward}} = 1.4 \times 10^{11} \text{ L mol}^{-1} \text{ s}^{-1}$	58
7	$\text{HO}_2 \rightleftharpoons \text{O}_2^- + \text{H}^+$	$k_{\text{backward}} = 5.0 \times 10^{10} \text{ L mol}^{-1} \text{ s}^{-1}$	58
8	$\text{cis-}\beta\text{-IEPOX} + \text{OH} \rightarrow \text{DHBO} + \text{CO} + \text{OH}$	$6 \times 10^{-3} \pm 13\%$	<i>a</i>
9	$\text{cis-}\beta\text{-IEPOX} + \text{OH} \rightarrow \text{DHMP} + \text{CO} + \text{OH}$	$35.4 \pm 11.9\%$	<i>a</i>
10	$\text{cis-}\beta\text{-IEPOX} + \text{OH} \rightarrow \text{methylglyoxal} + \text{glycolaldehyde} + \text{HO}_2$	$13.8 \pm 0.8\%$	<i>a</i>
11	$\text{cis-}\beta\text{-IEPOX} + \text{OH} \rightarrow \text{glyoxal} + \text{hydroxyacetone} + \text{HO}_2$	$6.3 \pm 1.0\%$	<i>a</i>
12	$\text{cis-}\beta\text{-IEPOX} + \text{OH} \rightarrow \text{IEPOXO} + \text{HO}_2$	$25.9 \pm 4.2\%$	<i>a</i>
13	$\text{cis-}\beta\text{-IEPOX} + \text{OH} \rightarrow \text{HBDO} + \text{HO}_2$	$18.6 \pm 0.9\%$	<i>a</i>
14	$\text{cis-}\beta\text{-IEPOX} + \text{OH} \rightarrow \text{hydroxyacetone} + \text{radical fragment}$	$2 \times 10^{-4} \pm 2 \times 10^{-2}\%$	<i>a</i>
sum	$\text{cis-}\beta\text{-IEPOX} + \text{OH} \rightarrow \text{product}$	$(1.0 \pm 0.3) \times 10^9 \text{ L mol}^{-1} \text{ s}^{-1}$	<i>b</i>
15	$\text{DHBO} + \text{OH} \rightarrow \text{hydroxyacetone} + \text{HO}_2$	$7.9 \times 10^8 \text{ L mol}^{-1} \text{ s}^{-1}$	34
16	$\text{DHMP} + \text{OH} \rightarrow \text{methylglyoxal} + \text{HO}_2$	$2.0 \times 10^8 \text{ L mol}^{-1} \text{ s}^{-1}$	34
17	$\text{DHMP} + \text{OH} \rightarrow \text{hydroxyacetone} + \text{HO}_2$	$1.1 \times 10^9 \text{ L mol}^{-1} \text{ s}^{-1}$	34
18	$\text{glycolaldehyde} + \text{OH} \rightarrow \text{glyoxal} + \text{HO}_2$	$1.2 \times 10^9 \text{ L mol}^{-1} \text{ s}^{-1}$	59
19	$\text{hydroxyacetone} + \text{OH} \rightarrow \text{methylglyoxal} + \text{HO}_2$	$1.1 \times 10^9 \text{ L mol}^{-1} \text{ s}^{-1}$	48
20	$\text{glyoxal} + \text{OH} \rightarrow \text{glyoxylic acid} + \text{HO}_2$	$9.2 \times 10^8 \text{ L mol}^{-1} \text{ s}^{-1}$	60
21	$\text{methylglyoxal} + \text{OH} \rightarrow \text{pyruvic acid} + \text{HO}_2$	$5.3 \times 10^8 \text{ L mol}^{-1} \text{ s}^{-1}$	61
22	$\text{methylglyoxal} + \text{H}_2\text{O}_2 \rightarrow \text{acetic acid}$	$1.7 \times 10^{-4} \text{ L mol}^{-1} \text{ s}^{-1}$	<i>c</i>
22	$\text{HBDO} + \text{H}_2\text{O}_2 \rightarrow \text{acetic acid}$	$0.48 \text{ L mol}^{-1} \text{ s}^{-1}$	<i>c</i>
23	$\text{pyruvic acid} + \text{H}_2\text{O}_2 \rightarrow \text{acetic acid}$	$0.75 \text{ L mol}^{-1} \text{ s}^{-1}$	53

^aEstimated partial rate constants of the single pathways of the OH-driven *cis-β*-IEPOX oxidation, describing the branching ratio, in the aqueous phase by the current study. ^bEstimated sum rate constant of the OH-driven *cis-β*-IEPOX oxidation in the aqueous phase by the current study.

^cEstimated based on ref 53.

**Figure 4.** Experimental (dots) and modeled (line) concentration-time-profiles of *cis-β*-IEPOX (left panel) and the quantified products (right panel).

no regards as to whether these are aldehydes, ketones, or carboxylic acids.^{52–54}

Based on the summarized reactions and the literature kinetic data (cf. Table 2), the second-order rate constants of the potential reaction pathways of *cis-β*-IEPOX were estimated based on the experimental findings using COPASI 4.19.⁶² Figure 4 shows the experimental concentrations and the model results having the best fit to the experimental data.

The modeled concentration-time-profile of *cis-β*-IEPOX fits well to the experimental results. The estimated second-order rate constants of the single pathways of the *cis-β*-IEPOX oxidation lead to an overall second-order rate constant of $1.0 \times 10^9 \text{ L mol}^{-1} \text{ s}^{-1}$. Compared to the experimental second-order rate constant of $(1.4 \pm 0.1) \times 10^9 \text{ L mol}^{-1} \text{ s}^{-1}$, the modeled second-order rate constant is 27% smaller than the

experimentally determined one. The discrepancy can be caused by missing sink processes for hydroxyl radicals in the model leading to a higher OH budget for the *cis-β*-IEPOX + OH reactions and consequently a smaller second order rate constant. The reaction rate constants of the reactions of *cis-β*-IEPOX and OH (Table 2, entry 8–14) were set to be free in order to obtain the best fit for the concentration time profiles for all experimentally observed products. In evaluation of the estimated branching ratios, it turned out that the reactions 8 and 14 are negligible for the modeling, which is in agreement with the gas-phase oxidation scheme published by Jacobs et al.⁵⁰

Having the kinetics and the product distribution with the good recovery, the importance of this single process has to be

addressed and compared to other atmospheric processes described in the literature.^{27,28,32}

Atmospheric Relevance. In order to assess the impact of the aqueous-phase reaction of *cis*- β -IEPOX toward hydroxyl radicals, a flux analysis of the pathways described was performed. As oxidation pathways, the reactions with hydroxyl radicals in the gas and aqueous phases (Table 3, entry 1, 5)

Table 3. Considered Reactions and Their Reaction Rate Constants for the Atmospheric Impact Estimation of the Aqueous-Phase OH Radical Reaction of *cis*- β -IEPOX

no.	pathway	reaction rate constant	references
Aqueous Phase			
1	OH radical reaction	$1.4 \times 10^9 \text{ L mol}^{-1} \text{ s}^{-1}$	this study
2	direct hydrolysis	$3.0 \times 10^{-2} \text{ L mol}^{-1} \text{ s}^{-1}$	27,28
3	SO ₄ ²⁻ assisted hydrolysis	$6.1 \times 10^{-3} \text{ L}^2 \text{ mol}^{-2} \text{ s}^{-1}$	27,28
4	NO ₃ ⁻ assisted hydrolysis	$3.7 \times 10^{-3} \text{ L}^2 \text{ mol}^{-2} \text{ s}^{-1}$	27,28
Gas Phase			
5	OH radical reaction	$1.52 \times 10^{-11} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$	32

were considered. Competing processing of IEPOX in general is described by the hydrolysis of the epoxide ring without nucleophiles and with sulfate and nitrate as attacking nucleophiles (Table 3, entry 2–4). Reaction rate constants of reactions 1 the reactions (entry 2–4) were estimated by the overall rate constants and branching ratios given in the literature.^{27,28} Here, the ring opening of IEPOX was determined by dedicated laboratory experiments using nuclear magnetic resonance techniques.

In order to estimate an upper limit, two scenarios were set up with cloud and non-cloud subscenarios which were merged together with a ratio of 0.15–0.85 cloud to non-cloud conditions according to Pruppacher and Jaenicke.⁶³ Scenario 1 has a liquid water content (LWC) of 0.25 g m^{-3} during the cloud phase and an aerosol LWC of $50 \mu\text{g m}^{-3}$ during the non-cloud phase. The second scenario has a LWC of 0.05 g m^{-3} and $10 \mu\text{g m}^{-3}$, respectively. The chosen conditions are in agreement with reviewed values given by Herrmann et al.⁶⁴ Subsequently, the aqueous-phase and gas-phase fractions were calculated by III

$$X_{A_{\text{aq}}} = \frac{C_{A_{\text{aq}}}}{(C_{A_{\text{aq}}} + C_{A_{\text{air}}})} \quad (\text{III})$$

$$= \frac{K_{\text{H(A)}}^* \times R^* \times T \times \text{LWC} \times 10^{-6}}{1 + K_{\text{H(A)}}^* \times R^* \times T \times \text{LWC} \times 10^{-6}}$$

The aqueous-phase hydroxyl radical concentrations are modeled mean values of model runs of the chemical aqueous phase radical mechanism (CAPRAM) 3.0i mechanism for a remote case.³⁸ The applied aerosol sulfate and nitrate concentrations are the representative values for the mid-Europe background as they were the measured values at the Melpitz sampling site, a rural background site, by Scheinhardt et al.⁶⁵ These authors published upper limit concentrations of $4.1 \times 10^5 \text{ mg L}^{-1}$ (2.5 mol L^{-1} calculated with a LWC of $10 \mu\text{g m}^{-3}$) for sulfate and $3.4 \times 10^5 \text{ mg L}^{-1}$ (3.5 mol L^{-1} calculated with a LWC of $10 \mu\text{g m}^{-3}$) for nitrate. The respective values for the non-cloud conditions in scenario 1 are scaled by the reciprocal factor of the LWC increase. Cloud sulfate and nitrate concentrations were taken from Ervens et al.⁶⁶ describing initial concentrations for the CAPRAM. The mean gas-phase hydroxyl radical concentration is estimated to be $1 \times 10^6 \text{ molec cm}^{-3}$ by Prinn et al.⁶⁷

Based on the constraints given in Table 4, the first-order rate constants of each pathway are calculated for pH = 0–6 and the four subscenarios, and their percentaged turnovers are displayed in Figure S14 for scenario 1 and Figure S15 for scenario 2. In order to calculate general turnovers, the percentaged values given in Figures S14 and S15 were averaged over pH = 0 to pH = 3 for the aerosol subscenarios and over pH = 4 to pH = 6 as these are the generally accepted ranges. The resulting overall percentaged turnovers are shown in Figure 5, disregarding the formed products and their subsequent reactions.

Scenario 1 with more diluted solutions is dominated by the aqueous-phase direct hydrolysis pathway with 56.6% which is mainly occurring in the non-cloud subscenario, where the aqueous-phase hydroxyl radical reaction pathway contributes with 29%. If a cloud is present, 96% of *cis*- β -IEPOX will be processed via the aqueous-phase oxidation pathway by hydroxyl radicals and with 3.4% by the aqueous-phase direct hydrolysis pathway. The pathways of the sulfate-assisted and nitrate-assisted reactions are minor in this scenario with about 5% each. In this scenario, the aqueous-phase hydroxyl radical oxidation pathway contributes with approximately 30% to the overall processing of *cis*- β -IEPOX. In a sensitivity study of scenario 1, the hydroxyl radical concentrations are decreased by 1 order of magnitude, leading to a processing totally dominated by the aqueous-phase direct hydrolysis pathways (77%) where the aq *cis*- β -IEPOX + OH reaction pathway has only a percentage of 4%.

In scenario 2 with more concentrated solutions, the sulfate-assisted pathway and nitrate-assisted pathway increase to contributions of 18.2 and 15.4% while the contribution of the direct hydrolysis pathway decreases by approximately 20–35.7%. Because of the decreased partitioning of *cis*- β -IEPOX to

Table 4. Constraining Parameters and Concentrations for the Atmospheric Impact Estimation

parameter	scenario 1		scenario 2	
	cloud conditions	non-cloud conditions	cloud conditions	non-cloud conditions
LWC	0.25 g m^{-3}	$50 \mu\text{g m}^{-3}$	0.05 g m^{-3}	$10 \mu\text{g m}^{-3}$
X_{aq}	0.994	0.034	0.972	0.007
X_{gas}	0.006	0.966	0.028	0.993
$[\text{OH}]_{\text{aq}}$	$2.2 \times 10^{-14} \text{ mol L}^{-1}$	$3 \times 10^{-12} \text{ mol L}^{-1}$	$2.2 \times 10^{-14} \text{ mol L}^{-1}$	$3 \times 10^{-12} \text{ mol L}^{-1}$
$[\text{SO}_4^{2-}]$	$3 \times 10^{-5} \text{ mol L}^{-1}$	0.5 mol L^{-1}	$6 \times 10^{-4} \text{ mol L}^{-1}$	2.5 mol L^{-1}
$[\text{NO}_3^-]$	$3 \times 10^{-5} \text{ mol L}^{-1}$	0.7 mol L^{-1}	$6 \times 10^{-4} \text{ mol L}^{-1}$	3.5 mol L^{-1}
$[\text{OH}]_{\text{gas}}$	$1 \times 10^6 \text{ molecule cm}^{-3}$	$1 \times 10^6 \text{ molecule cm}^{-3}$	$1 \times 10^6 \text{ molecule cm}^{-3}$	$1 \times 10^6 \text{ molecule cm}^{-3}$

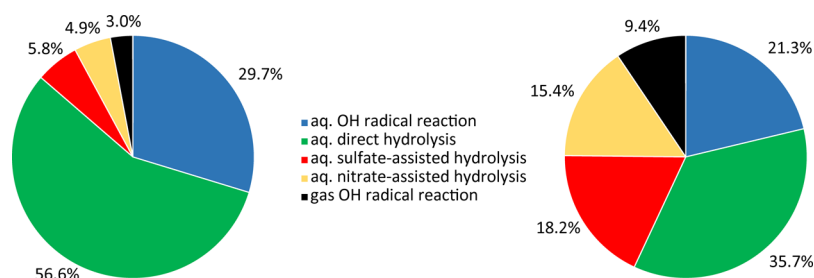


Figure 5. Relative turnovers of the five initial pathways of the *cis*- β -IEPOX processing in the estimation scenario 1 (left) and scenario 2 (right).

the aqueous phase, the contribution of the gas-phase oxidation reaction pathway rises to 9.4%. The contribution of the aqueous-phase OH radical reaction pathway is still approximately 21%. Performing the same sensitivity study on the hydroxyl radical concentration, the contributions of the direct hydrolysis pathway (49.3%), sulfate-assisted hydrolysis pathway (25.2%), and nitrate-assisted hydrolysis pathway (21.3%) increase strongly. The oxidation reactions show only 1.3% contribution for the gas-phase oxidation reaction and 2.9% for the aqueous-phase oxidation.

Overall, the aqueous-phase oxidation of *cis*- β -IEPOX by hydroxyl radicals investigated in the present study contributes in a range of 29.7–21.3% to the entire atmospheric *cis*- β -IEPOX reactions, and the authors highly recommend the consideration of this reaction pathway in atmospheric models.

CONCLUSION

Within the present study, the aqueous-phase second-order rate constants for the reaction of *cis*- β -IEPOX with hydroxyl radicals were determined for the temperature range 278–318 K to be in the range of 1.1×10^9 to 2.0×10^9 L mol⁻¹ s⁻¹. These constants are in good agreement with the kinetic data of the mentioned structurally similar compounds. For the product studies for the OH-driven oxidation reaction, short chain carbonyls and acids such as glycolaldehyde (maximum molar fraction of 6.9%), hydroxyacetone (7.4%), methylglyoxal (10.3%) and glyoxal (6.6%) as well as acetic acid (13.9%) were observed as formed products. Based on these experimental findings together with known kinetic data from the literature a box model was created to understand the presented reaction mechanism for the aqueous phase. The model estimates a well-fitting reaction rate constant for the reaction of *cis*- β -IEPOX with hydroxyl radicals to be 1.1×10^9 L mol⁻¹ s⁻¹ at 298 K which is 27% lower compared to the experimental determined rate constant ($k_{2nd} = (1.4 \pm 0.1) \times 10^9$ L mol⁻¹ s⁻¹).

The overall relative sink contribution of the OH-driven oxidation reaction is found to be in the range of 21–30% related to the overall atmospheric processing in both the gas and in the aerosol particle phase based on available kinetic data. In conclusion, it needs to be considered as an important pathway of the atmospheric *cis*- β -IEPOX processing.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpca.9b08836>.

Aqueous-phase partitioning plot of *cis*- β -IEPOX, lamp spectrum, temperature-dependent second-order rate constants, estimation of the second-order rate constant

using SAR method, concentration time profiles, signal time profiles of unassigned *m/z*, turnover plots of the subscenarios, table of quantified analytes, and ¹H NMR spectrum of *cis*- β -IEPOX (PDF)

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Notes

The authors declare no competing financial interest.

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